Project Report

on

AI-Driven Traffic Signal Optimization Using CCTV Feeds

Submitted in partial fulfillment of requirements for the award of the degree

Bachelor of Technology
In
Computer Science and Engineering
To
IKG Punjab Technical University, Jalandhar

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Batch: 2023-2027

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CANDIDATE’S DECLARATION

I hereby declare that the work which is being presented in the 6th semester, entitled **“AI-Driven Traffic Signal Optimization Using CCTV Feeds”,** in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering and submitted to CGC College of Engineering, Landran, Mohali is an original piece of work carried out by me during the period from January 2026 to May 2026 under the supervision of **Ms. Neha Sharma**, department of CSE, CGC College of Engineering, Landran, Mohali, Punjab, India.

The matter embodied in this File has not been submitted by me for the award of any other degree of any other University/Institute.

Dhruv
Candidate Name & Signature

The File entitled **AI-Driven Traffic Signal Optimization Using CCTV Feeds** submitted by **Dhruv** to Department of CSE for the award of the degree of Bachelor of Technology is a Bonafide record of the project work carried out by her under my supervision and guidance. This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Ms. Neha Sharma
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Certificate

I hereby declare that the 6th Semester Training Report entitled ("AI-Driven Traffic Signal Optimization Using CCTV Feeds") is an authentic record of my own work as requirements of 6th semester academic during the period from January 2026 to June 2026 for the award of degree of B.Tech. (**Computer Science & Engineering**), CGC-College of Engineering, Landran, Mohali under the guidance of **Ms. Neha Sharma**

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Certified that the above statement made by the student is correct to the best of our knowledge and belief.

Signatures

Examined by:

1. 2. 3. 4.

Head of Department
(Signature and Seal)

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Finally, indebtedness is expressed to all individuals who have directly or indirectly contributed to the successful completion of this 6th Semester Training.

Dhruv
(Signature)

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
CV	Computer Vision
DL	Deep Learning
YOLO	You Only Look Once
ROI	Region Of Interest
FPS	Frames Per Second
API	Application Programming Interface
UI	User Interface
CCTV	Closed Circuit Televiosion
ML	Machine Learning
RGB	Red Green blue
OpenCV	Open Source Computer Vision Library
NMS	Non-Maximum Suppression
IoT	Internet Of Things
GPU	Graphics Processing Unit

ABSTRACT

Traffic congestion is one of the biggest issues in the contemporary urban regions, which causes longer travelling time, overuse of fuel, and pollution of the environment. The conventional traffic light systems utilize a predetermined time span and fail to adjust with the actual traffic patterns, leading to poor traffic control. This project suggests an intelligent traffic signal system as an AI-based solution, which is dynamically adjusted in terms of the traffic density of the vehicles on the road.

The system applies the YOLOv8 (You Only Look Once) object detector made up of deep learning to process traffic video. It reads and recognizes various kinds of vehicles including cars, motorcycles, buses, and trucks. The system also uses unique IDs to track vehicles so that the same vehicle is not counted more than once to enhance accuracy. It employs a weighted traffic calculation methodology, in which the various types of vehicles can contribute differently to the total traffic load based on their size and influence on road space.

According to the traffic density calculated, the system categorizes the traffic conditions as low, medium, or heavy congestion. Based on this, it dynamically varies the traffic signal timing so as to maximize the vehicle flow. A region of interest (ROI) mechanism is also inbuilt into the system to limit it to only the relevant areas of the road, and to prevent unnecessary detections. Also, smoothing is used to stabilize and maintain consistency in the signal decisions without abrupt changes.

The project runs on Python, OpenCV, and the Ultralytics YOLO framework and offers a real-time visual simulation of traffic signals and detection of vehicles, traffic density, and information on the timing of traffic lights. The system shows that computer vision and artificial intelligence can be successfully implemented to smart traffic control.

Though the system is now being developed in the form of a software prototype, it has high potentials of implementation in the real world. It can be expanded by incorporating with hardware components like microcontrollers, cameras, and traffic signal controllers to form a complete automated smart traffic system. This project emphasizes a pragmatic solution to traffic congestion and better transportation efficiency in urban areas with the modern AI technologies.

Chapter – 1

INTRODUCTION

1.1 Overview

Modern cities face serious problems related to the growing density of traffic due to high population growth rates and a large number of vehicles used by city residents. Effective traffic management is difficult due to a lack of responsiveness of traditional traffic systems to real-time road conditions. This leads to increased waiting time for motorists, fuel waste, and, as a consequence, increases air pollution.

Most of today's traffic light controllers are based on a timed cycle. They do not take into account the number of vehicles currently in the corresponding lane. Consequently, there will be situations when certain lanes will be completely unused while others will suffer from high congestion.

Due to the development of technologies such as Artificial Intelligence (AI) and Computer Vision, there is a possibility of creating intelligent systems for monitoring and managing traffic flows. The current project aims to design an intelligent traffic light system that works by analyzing video images of lanes.

The proposed solution will significantly exceed the capabilities of ordinary systems due to working in multi-lane traffic environment. It will be able to monitor several roads at the same time and guarantee the issuance of a green signal for just one lane.

1.2 Background

Recent advancements in machine vision and deep learning have enhanced the ability of machines to analyze visual input data. Advanced object detectors, like YOLO (You Only Look Once), are able to detect multiple objects in video feed.

In traffic management, cameras are beginning to replace the conventional sensor technology because they generate more information and have wider coverage, enabling them to be used for real-time analysis.

However, currently available methods do not fully exploit the potential offered by artificial intelligence technologies for dynamically managing traffic signals. Moreover, problems associated with heavy traffic, overlapping cars, and an unbalanced distribution of traffic signals are yet to be solved. This project focuses on these problems.

1.3 Motivation

The inspiration for this project comes from real-world traffic issues. Cars frequently sit idle at signals despite being no cars on the road, whereas highly busy lanes get little to no signal time.

A third factor to consider is the impact on the environment. The result of heavy traffic is too much fuel burning and excessive emissions, leading to pollution.

Finally, current traffic systems have issues of unfairness. In some artificial intelligence systems, the most crowded lane keeps getting preference, leaving other lanes waiting for long periods.

This project will work towards developing an intelligent traffic system that:

- adjusts according to traffic flows
- allocates signal time equally across all lanes
- increases traffic flow efficiency

1.4 Problem Statement

The major problem that this project attempts to address is that of inefficient traffic signal systems that cannot dynamically adjust themselves according to the actual traffic situation.

Other problems that need to be considered are:

- imprecise measurement of the number of cars when there is heavy traffic because of overlapping
- unequal distribution of traffic lights favouring one lane over another

- lack of integration among several lanes at the intersection

None of the existing systems can:

- measure the number of lanes at once through visual data
- calculate congestion even in heavy traffic
- distribute the traffic lights efficiently
- adjust traffic lights dynamically

1.5 Scope of the Project

In this project, the aim was to design an intelligent traffic signal system with multiple lanes based on video analysis.

The scope is as follows:

- Real-time vehicle detection using deep learning
- Estimation of traffic density based on weighted vehicle count
- Estimation of congestion based on occupancy ratio
- Multi-lane signal coordination
- Fair distribution of signal based on waiting time and priority criteria
- Simulation of traffic signals using videos

This was simulated using software. But the same can be made applicable in real life through camera, microcontroller, and traffic signals.

1.6 Report Structure

The structure of this report enables the reader to have a clear picture about the project at hand. The report starts with an introduction to the problem under consideration and its context. This is followed by a thorough description of the problems that exist and the requirement of a better system. Further, the technologies associated with the subject are discussed along with the objectives and viability of the proposal. Finally, the method of execution and results of the system are described.

Chapter – 2

PROBLEM STATEMENT

2.1 Introduction

Cities today are packed with cars, and keeping traffic moving smoothly is a real challenge. As the number of vehicles keeps climbing, basic traffic signals just can't keep up. They don't know how to handle the constant changes on the road, so we end up with unnecessary traffic jams, longer delays, and roads that aren't being used as well as they could be.

This chapter dives into what's wrong with the traffic systems we have and why cities need smarter, more responsive solutions.

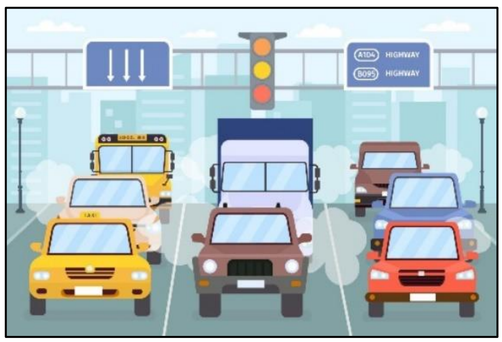

Fig 2.1 Traffic Congestion

2.2 Existing System

Most current traffic signals run on fixed timers. They don't pay attention to what's actually happening out on the road — just the average traffic patterns the system was built for.

Here's how these systems typically work:

- Every lane gets a set green light time, no matter what's actually happening
- They ignore how many vehicles are waiting
- Signal lengths don't change for accidents, crowds, or empty roads

Sure, these systems are easy to set up, but they rarely work well. When rush hour hits or roads are empty, they can't adjust, so drivers end up frustrated and stuck.

2.3 Limitations of Current System

Traditional traffic lights have a lot of problems:

- They don’t adapt to real-time traffic
- Cars sit idling at red lights even if the street is empty
- Congested lanes don’t get enough green time
- Engines running longer means burning unnecessary fuel
- Extra pollution follows
- Some setups still need a person to manually change the lights
- Occlusion problem (vehicles overlapping)
- Inaccurate counting in dense traffic
- Unfair lane priority

All of this adds up to old-fashioned systems that just aren’t efficient for today’s traffic.

2.4 Problem Definition

The heart of the problem is that current traffic signal management simply isn’t smart enough — there’s no real-time analysis going on.

There’s no widely used system that can:

- Watch traffic live with cameras
- Detect and track vehicles accurately
- Measure how crowded each lane is, right now
- Change light timing on the fly based on actual road conditions

Because of this, traffic signals can’t keep cars moving efficiently. Instead, we get bottlenecks, wasted minutes, and a lot of frustration at red lights.

2.5 Proposed Solution

To fix these issues, this project introduces an AI-powered smart traffic signal system.

Imagine a system that:

- Uses computer vision and deep learning to spot cars
- Checks how busy the roads are in real time
- Sorts traffic into categories by overall load
- Adjusts green light time for each lane based on what’s truly needed
- Multi-lane input (4 videos)
- Single green signal coordination
- Waiting-time fairness
- Occupancy-based congestion

With this approach, traffic flows better, people spend less time waiting, and roads work more efficiently.

We’re starting with a software simulation, but this design could easily move into the real world with the right hardware.

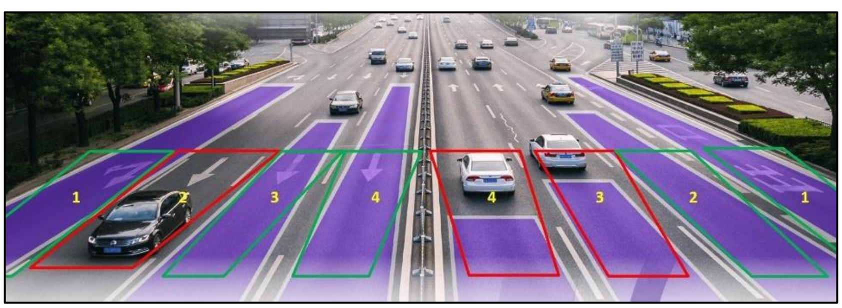

Fig 2.2 No. Of Vehicles Detected

2.6 Project Objectives

Here’s what this project sets out to do:

- Build a real-time vehicle detection system

- Use AI to figure out how crowded the roads are
- Design a traffic signal that adapts itself on the fly
- Reduce traffic jams and long waits
- Visually simulate these smarter traffic signals for testing and improvement.
- Multi-lane analysis
- Fairness logic
- Occupancy estimation
- Realistic signal simulation

Chapter – 3

SURVEY OF TECHNOLOGY

3.1 Introduction

As technology has moved forward, the way we manage traffic has really changed over the years. We've come up with many different strategies to help monitor and control traffic, with each offering varying levels of effectiveness. These range from older manual setups all the way to modern automatic and smart systems. In this chapter, we'll explain the technology that goes into traffic management and also describe the specific methods we used for this study.

Our main goal in this chapter is to understand how tools like Artificial Intelligence, Computer Vision, and Deep Learning can improve traffic signal management systems.

3.2 Traditional Traffic Management Systems

For a long time, traffic control systems mostly relied on traffic lights that ran on a set schedule or on people managing traffic by hand.

3.2.1 Fixed-Time Signal Systems

With fixed-time systems, traffic lights just stick to a schedule, no matter how much or how little traffic there actually is. They're straightforward to set up, but they're not very good at handling traffic because they can't react to what's happening right now.

3.2.2 Manual Traffic Control

Sometimes, traffic police direct traffic themselves. This way, they can be flexible and adjust to things on the spot, but it's just not practical for managing a lot of traffic all the time, and it really wears people out.

3.3 Sensor-Based Traffic Systems

To make traffic flow better, people started using systems that rely on sensors. These systems use things like:

- Inductive loop sensors
- Infrared sensors
- Ultrasonic sensors

Workers put these sensors right on the roads to sense when cars are there. Then, based on what the sensors pick up, they can change how long traffic lights stay green or red.

But these systems have their downsides:

- They cost a lot to put in and keep running.
- They also only cover a small area.
- And when the weather gets bad, they don't work as well.

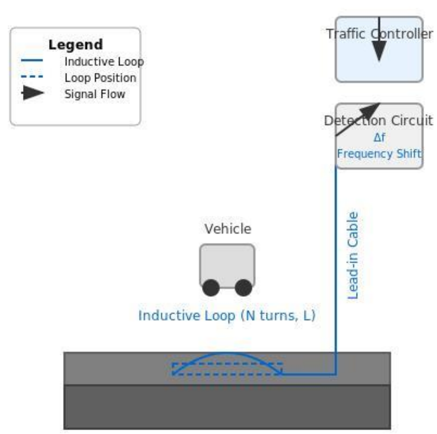

Fig 3.1 Sensor Based Traffic Monitoring System

3.4 Camera-Based Traffic Monitoring

These days, you see camera-based systems used all over for watching traffic. Cameras record live video, and then that video can be looked at to spot cars and figure out how much traffic there is.

Compared to sensor systems, cameras offer:

- Wider coverage
- Much more detailed information
- And are easier to set up in different places.

But just having cameras isn't enough to make smart choices unless something clever processes the video. That's where AI and Computer Vision really become important.

Fig 3.2 Camera Based Traffic Monitoring System

3.5 Artificial Intelligence in Traffic Management

Artificial Intelligence, or AI, has opened up lots of new ways to handle traffic. Systems that use AI can look at tons of information and then make decisions all on their own.

For traffic management, AI helps with things like:

- Finding vehicles

- Guessing what traffic will be like
- Making traffic lights work better
- And spotting patterns

These AI systems work better because they can learn and adjust as conditions change, getting smarter over time.

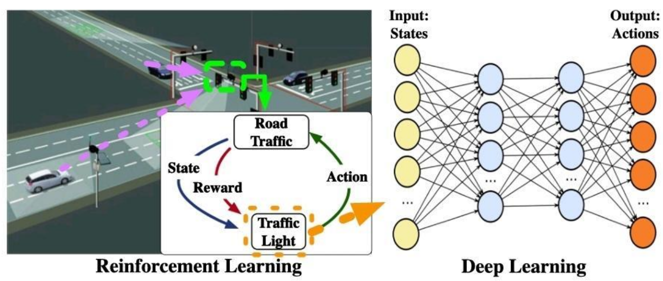

Fig 3.3 Artificial Intelligence in Traffic Management

3.6 Computer Vision Techniques

Computer Vision is part of AI that teaches computers how to 'see' and understand pictures and videos.

In this particular project, we're using computer vision to:

- Spot vehicles in video frames
- Figure out what kind of vehicles they are
- And keep an eye on where they're going

This means our system can understand what's happening with traffic right now without needing any physical sensors on the road.

3.7 YOLO (You Only Look Once) Algorithm

YOLO, which stands for 'You Only Look Once,' is a really popular algorithm for deep learning that can spot objects quickly, especially useful for things that need to happen in real-time.

What's great about YOLO is that it's:

- Superfast at finding things
- Pretty accurate
- And can even find many different objects in just one picture.

The way YOLO works is it breaks an image into smaller parts and then tries to guess where objects are by drawing boxes around them and saying what it thinks those objects are.

For this project, we're specifically using YOLOv8 to find vehicles such as cars, motorcycles, buses, and trucks in our traffic videos.

Fig 3.4 Vehicles Detection Using YOLO

3.8 OpenCV Library

OpenCV, which is short for Open Source Computer Vision Library,' helps with working on images and videos.

In our project, we use OpenCV for things like:

- Taking in video
- Working on each frame
- Drawing those boxes around detected objects
- And showing everything.

It gives us good tools to make computer vision applications that work in real-time.

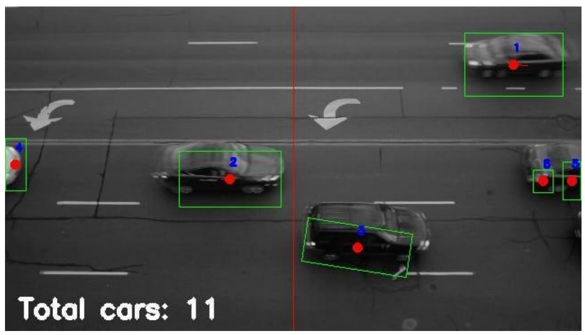

Fig 3.5 Box Creation Using OpenCV

3.9 Python Programming Language

We're using Python as the main programming language to build this system.

Python has a lot going for it:

- It's easy to pick up and use
- There are tons of libraries out there for AI and computer vision
- And there's a big, helpful community if you ever get stuck.

Python makes it simple to put YOLO and OpenCV together to create our traffic system.

3.10 Summary

So, in this chapter, we talked about all sorts of technologies used in managing traffic, going from the old ways to today's AI-driven solutions. We also went over the specific tools and methods we're using in our project, like YOLO, OpenCV, and Python.

Using Artificial Intelligence and Computer Vision gives us a much better and more adaptable way to control traffic lights, which is exactly what you need when building smart traffic systems.

Chapter-4

RATIONALE / NEED FOR THE PROJECT

4.1 Introduction

As the cities grow and the population increases, the number of visitors also seems to increase steadily. This creates a need for green space visitor monitoring systems. Our modern visitor control machinery fails to address the increasing complexity of roads that result in visitors, wasted gas and time. This chapter describes the need for intelligent visitor lighting fixtures in effective site visit management systems and their packages.

4.2 Need for Smart Traffic Systems

Traditional traffic signal systems operate on fixed timing and do not consider real-time traffic conditions. As a result, they fail to manage traffic efficiently.

In many situations:

- Vehicles wait unnecessarily at empty roads
- Congested lanes do not receive sufficient signal time
- Traffic flow becomes unbalanced

This creates a need for a system that can:

- analyze traffic dynamically
- respond to real-time conditions
- optimize signal timing automatically

A smart traffic system can overcome these limitations and provide better traffic control.

4.3 Need for Multi-Lane coordination

In practical cases of intersections, several lanes influence each other. In case of having more than one lane green at the same time, confusion may arise.

Thus, it is necessary that:

- only one lane gets green signal
- others remain stopped
- all the lanes are taken into consideration

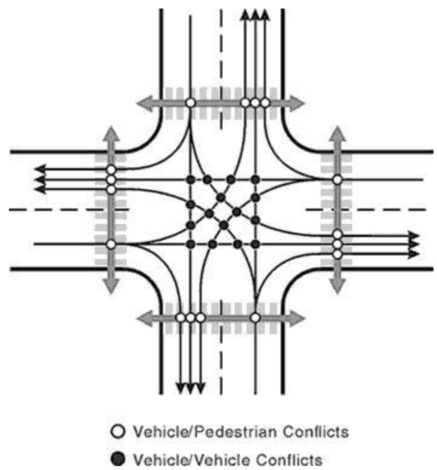

Fig 4.1 Traffic Signal in Multilane System

4.4 Role of Artificial Intelligence

Artificial Intelligence has a central position in traffic management now-a-days. It allows big data processing, decision making without human intervention.

See below for how AI is applied in this project:

- Detecting vehicles from video streams.
- Analyzing traffic d ensity.
- Adapting signal timings in real time to traffic scenarios.

  The AI system is more versatile and dynamic than traditional methods.

4.5 Importance of Computer Vision

The Computer Vision technique is an approach by which computers or machine learning algorithms can interpret the information provided through images and video feeds. Unlike traditional methods that require installing various sensors, computer vision allows traffic analysis via cameras.

There are several advantages of using computer vision:

- no requirement for costly sensor installations
- larger area coverage
- real-time traffic surveillance

In our project, we utilize computer vision techniques to detect vehicles in addition to assessing the traffic situation in different lanes.

4.6 Advantages of the Proposed System

The AI traffic signal control system has the following advantages:

- **Adaptive Signal Timing**: Signal timing is dependent on traffic flow
- **Coordination between multiple lanes**: One lane receives green signal at a time
- **Signal fairness**: Time in queue is used to avoid starvation of vehicles
- **Accurate congestion calculation**: Combines vehicle number and occupancy percentage
- **Waiting time minimization**: Eliminates unnecessary vehicle waiting
- **Effective traffic movement**: Balanced signal distribution between lanes
- **Cost-effectiveness**: Camera based system replaces expensive sensors
- **Real-time operation**: Works round-the-clock with real-time video feeds

These features make the system smarter than conventional traffic systems.

4.7 Application Areas

This system can find its applications in many fields like:

- city road crossings
- highways and multi-junction roads
- smart city technology
- traffic monitoring and control facilities

This system can be deployed practically through incorporation of hardware modules like traffic lights and embedded systems.

4.8 Summary

In this chapter, the importance of creating an intelligent traffic signal management system has been discussed. The drawbacks associated with conventional techniques have been pointed out. In addition to that, the benefits of implementing the technology of Artificial Intelligence and Computer Vision have been mentioned.

The traffic signal management system suggested here not only adjusts itself according to the traffic situation but also takes care of the fair allocation of signals in all the lanes simultaneously.

Chapter – 5

OBJECIVES

5.1 Introduction

This project's main objective is to come up with an intelligent traffic signal system using modern-day technologies like Artificial Intelligence and Computer Vision. Contrary to the conventional methods where the timing of signals is fixed, the objective of the current project is focused on analyzing the traffic in real time.

The proposed intelligent system will be capable of handling traffic for more than one lane at any particular instant. This chapter will address the major objectives of the project.

5.2 Primary Objectives

The main goals of the proposed project include the development of an effective and intelligent traffic management system which is capable of coping with actual traffic situations:

- Building a detection system that would enable detecting vehicles through video input
- Recognizing and categorizing different types of vehicles like cars, motorbikes, buses, and trucks
- Conducting traffic analysis for multiple lanes at once
- Developing a well-coordinated traffic light system in which only one lane will get the green light at any given time
- Adapting the timing of traffic lights according to actual traffic conditions
- Minimizing unnecessary wait times at junctions

5.3 Secondary Objectives

Aside from the objectives stated above, the objectives below shall be met to enhance the intelligence, fairness, and reality of the system:

- Introducing fairness in signal allocation through waiting time priority
- Avoiding lane starvation through balanced signal distribution
- Accurate congestion estimation based on vehicle number and lane occupancy
- Use of ROI for analysis in traffic
- Use of weighted values to consider effects of each vehicle in traffic
- Smoothing to ensure consistent decisions
- Reality simulation through stream pauses in lanes with red signals
- Performance enhancement through optimization, including frame skipping

5.4 Technical Objectives

Technical aspects of the project will involve the creation of a highly efficient and optimized system that utilizes the proper tools and techniques, including:

- Using YOLOv8 to detect vehicles in real time
- Using OpenCV to process videos, visualize information, and simulate the scenario
- Designing the system in Python
- Analyzing multiple video streams for each lane of the road
- Using the concept of occupancy to estimate congestion in dense traffic situations
- Optimizing the system to work effectively on a regular computer by minimizing the computational effort required

Fig 5.1 Technical Objectives of the System

5.5 Summary

This chapter outlined the major objectives of the project, covering both functional and technical aspects. The focus is on building an intelligent traffic signal system that can respond to real-time conditions using AI and Computer Vision. By doing so, the system aims to make traffic management more efficient, responsive, and practical for modern urban environments.

Chapter-6

FEASIBILITY STUDY

6.1 Introduction

Before implementing any system, it is crucial to assess its practicality and viability through feasibility studies. Such studies assist in evaluating technical, financial, and operational issues related to a particular system.

In this chapter, the feasibility of an AI-based smart traffic signal system will be examined. Such systems are characterized by multiple lane video inputs, intelligent decision making, and processing optimization.

Fig 6.1 Types of Feasibility

6.2 Technical Feasibility

Feasibility in technical terms means availability of technology and resources necessary to develop and implement the system.

In this project:

- The software will be implemented using the Python programming language which is popularly used for implementing AI and computer vision
- YOLOv8 will be used for real-time object detection with high precision
- OpenCV will be used for image processing, visualization, and simulation purposes
- The system will process multiple video inputs corresponding to each traffic lane
- Methods like frame skipping, occupancy estimation, etc. will be used to improve

efficiency

- The system is implemented on an average computer without special hardware

Though multiple lane processing introduces complexities, optimization techniques allow efficient operation of the system. Therefore, the proposed system is technically feasible

6.3 Economic Feasibility

The economic feasibility will decide whether the project is financially viable or not.

For the proposed project:

- The software tools that have been used (Python, OpenCV, YOLO) are open source and free
- Expensive hardware components are not needed
- Camera surveillance has been used rather than sensors
- It can be implemented using existing CCTV infrastructure

Hence, the project is economically feasible as compared to other systems where huge investments are needed.

6.4 Operational Feasibility

Operational feasibility relates to the ability of the system to work effectively in practical scenarios.

For the given project:

- The system is capable of analyzing traffic situations by means of video input in real-time.
- It facilitates multi-lane monitoring as well as coordination of traffic lights.
- It automatically chooses which lane receives green light depending on traffic condition.
- It takes account of waiting and traffic jams to maintain balance in traffic movement.
- It reduces human intervention in traffic management.

Therefore, the system can be considered operationally feasible due to its user-friendliness and integration capabilities within the traffic monitoring environment.

6.5 Practical Feasibility

Practical feasibility deals with the question of applying the system into real-life situations. Though the present approach uses software, it can be implemented in real-life situations through the use of:

- CCTV cameras for traffic monitoring in real-time,
- embedded systems/microcontrollers for controlling signals,
- and traffic signals for implementing the system in real life.

In addition, the system design that involves multi-lane coordination and intelligence makes the application feasible for use in smart cities. This means that the project has high practical feasibility.

6.6 Summary

This chapter analyzed the feasibility of the proposed system from technical, economic, operational, and practical perspectives. The results show that the project is feasible and can be successfully implemented using available resources

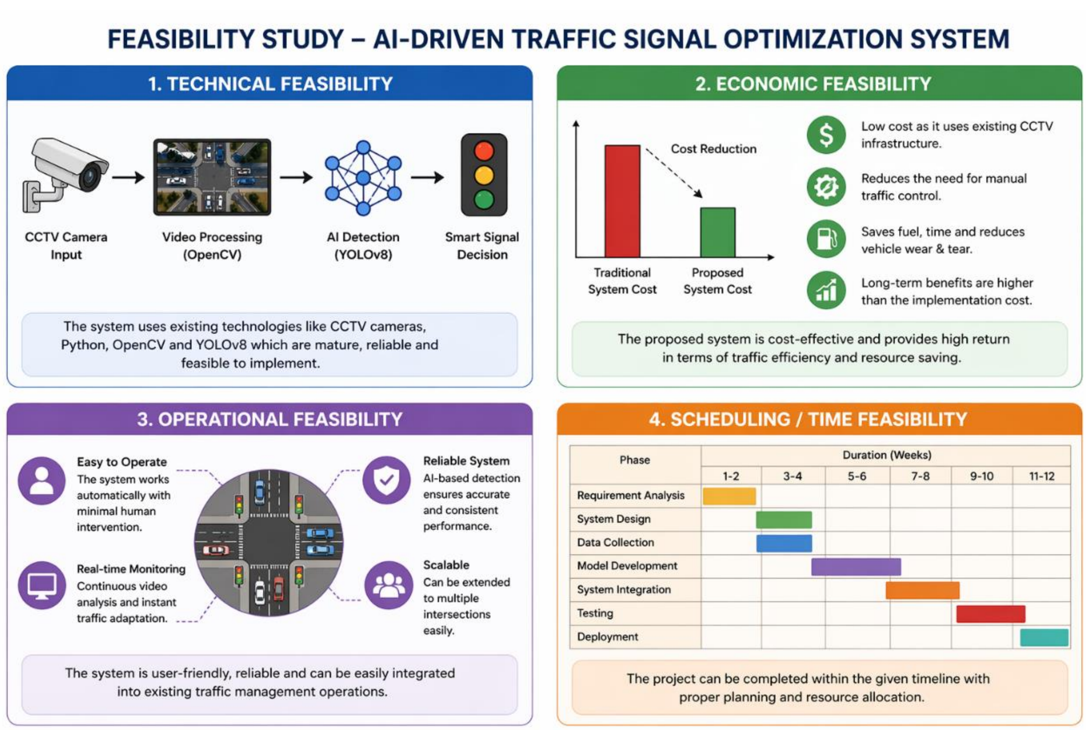

Fig 6.2 Feasibility Study Overview

Chapter – 7

METHODOLOGY

7.1 Introduction

This chapter provides details regarding the methodologies adopted for developing the intelligent traffic light system. The methodology adopted involves designing the system in such a manner that every component of the system has its own set of functionalities and plays an important role in managing traffic conditions.

The development of this project involves the utilization of Artificial Intelligence, Computer Vision, and real-time decision-making techniques.

Methodology primarily includes:

- system architecture
- working modules
- technologies and tools used

7.2 System Architecture

This suggested system will have a modular structure where several modules will collaborate to conduct traffic analysis and control signals.

The modules involved in the system include:

- Module of Input (Video Source):

  Here, several video sources will be used to represent different lanes of the traffic.
- Processing Module:

  This is the module that will process video frame images and use AI models for vehicle detection.
- Analysis Module:

  This module will determine the traffic volume, occupancy, and congestion in each lane of the traffic.
- Decision Module:

  This module will decide which lane needs to have the green light signal.
- Output Module:

  This module displays the results.

7.3 Components of the System

7.3.1 Input Module

Several video files are used to create realistic traffic scenarios.

- Video files are processed frame by frame.
- The dimensions of the frames are adjusted to increase processing speed.
- Only relevant frames are processed.

7.3.2 Vehicle Detection Module

The vehicle detection process is conducted via the YOLOv8 model.

- Identification of different types of vehicles such as motorcycles, cars, buses, and trucks.
- Bounding boxes are drawn on identified vehicles.
- Classification of detected vehicles by type.

This component is vital to traffic analysis.

7.3.3 Region of Interest (ROI)

A region of interest is established to enhance accuracy.

- Only vehicles within the defined ROI are analyzed.
- All other areas irrelevant to the road are excluded.
- Furthermore, it minimizes errors in vehicle detection.

7.3.4 Traffic Analysis Module

The traffic load is determined through:

- The total number of vehicles detected.
- The weight of each vehicle.

Each vehicle has a certain weight value:

- A motorcycle receives the lowest weight.
- Cars have medium weights.
- Buses and trucks have the highest weights.

The total traffic load is computed by summing all weights.

7.3.5 Occupancy Estimation

In dense traffic, not all the vehicles are detected because of overlapping.

To counter that,

- Occupied area is calculated
- Occupied area compared to total road area gives occupancy ratio

This makes detection more efficient for dense traffic.

7.3.6 Multi-Lane Processing

The system takes care of multiple lanes at the same time.

The system will calculate for every single lane:

- Number of vehicles
- Traffic load
- Occupancy
- Congestion level

With that, the lanes can be compared.

7.3.7 Signal Decision Module

This module is the most crucial part of the system.

The basis of signal decisions is,

- Traffic load
- Occupancy ratio
- Traffic waiting time

Prioritization is done for every lane and the lane having the most priority will have the green signal.

7.3.8 Fairness Mechanism

So that one lane doesn't take up all the time,

- Traffic waiting times are noted down
- Lanes which waited longer get priority
- Cooldown after the green signal

7.3.9 Signal Control Logic

There are only three states in the signal system:

- Green - Vehicles allowed through
- Yellow - Transition state
- Red - Vehicles not allowed

Green signal happens only in one lane at any point.

7.3.10 Video Simulation (Freeze Logic)

To simulate realistic traffic light scenarios,

- The video of green lane moves as it is
- The red lane's video is frozen
- The yellow lane is moved very slowly

7.4 Tools and Technologies Employed

Tools and technologies employed for the implementation of the system include:

- Python - programming language
- YOLOv8 - vehicle detection
- OpenCV - video processing
- NumPy - numerical operations

7.5 Data Flow in the System

The process flow in the system is as follows:

are inputted

1. Frames are extracted from videos
2. Vehicle detection is carried out
3. Traffic loads are computed
4. Lane priority is established
5. Signal priority is decided
6. Results are outputted
7. The process starts again

7.6 Working Process

The system works consistently in real-time mode:

- traffic analysis in each lane is done
- congestion level comparison is carried out
- the most congested lane is selected
- the signal is updated periodically
- fairness among lanes is ensured

The above loop is repeated till the system is stopped.

7.7 Conclusion

This chapter discussed the methodology involved in the development of the project. The smart traffic signal system integrates computer vision, traffic analysis, and intelligent signal prioritization algorithms to achieve its purpose.

Multi-lane traffic analysis, traffic occupancy calculation, and lane fairness consideration make the system more efficient and practical.

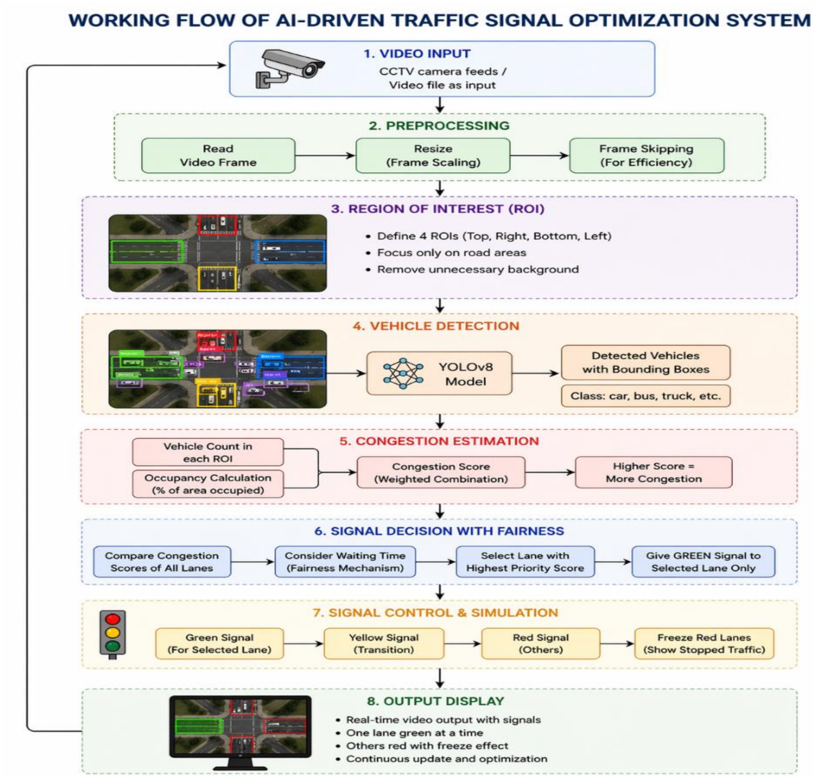

Fig 7.1 Methodology of Proposed System

Chapter – 8

PROJECT IMPLEMENTATION PLAN / TIMELINE

8.1 Introduction

Effective planning plays a critical role in the success of any given project. The implementation plan assists in organizing work and managing time to ensure efficiency in each stage of the project.

The current chapter explains the stages and timeline observed throughout the development process of the artificial intelligence traffic control system. Several stages were involved in the entire process of development, from problem comprehension through testing and implementation.

8.2 Development Phases

System development proceeded through various stages in order to ensure its consistent progress and proper implementation.

8.2.1 Phase 1: Problem Identification

In the first stage, the problem statement was conducted, which consisted in identifying issues associated with traffic congestion and poor performance of traditional traffic systems.

Activities carried out:

- analysis of limitations of fixed time traffic signals
- need for traffic control system
- problems with multi-lane traffic control

8.2.2 Phase 2: Requirement Analysis

In this stage, requirements were identified.

Activities carried out:

- gathering information on required tools and technologies (Python, OpenCV, YOLO)
- identifying input/output requirements
- identification of objectives and system scope

8.2.3 Phase 3: System Design

At this stage, system design was elaborated.

Activities carried out:

- system architecture development
- module design (detection, analysis, decision)
- logic for processing multi-lane traffic flow

8.2.4 Phase 4: Implementation

At this stage, the system was coded.

Activities carried out:

- implementation of YOLOv8 model for detection purposes
- ROI-based detection algorithm implementation
- calculation of traffic load algorithm
- multi-lane traffic processing algorithm development
- signal decision logic development
- traffic load fairness mechanism implementation

8.2.5 Phase 5: Testing and Optimization

Following the implementation process, testing was done, and the system was optimized.

Activities Undertaken:

- Traffic video testing with varying types
- Detection error identification
- Achieving accuracy through occupancy calculation
- Skipping frames to enhance system efficiency
- Resolving slow processing challenges

8.2.6 Phase 6: Final Integration and Display of Outputs

In this last phase, all modules were integrated, and outputs were presented.

Activities Undertaken:

- Integration of all lanes in one module
- Visualization of signals

- Inclusion of video freeze for realistic presentation
- Output display (count and signal status)

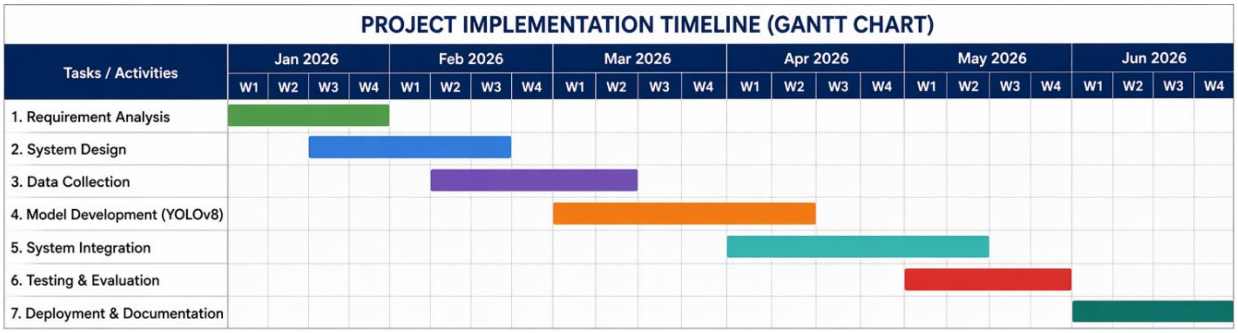

Fig 8.1 Gantt Chart Showing Project Timeline

8.4 Development Strategy

In the course of the project, there was an incremental development strategy where the completion of one phase was necessary before the start of another phase. There was testing and improvement throughout the process to achieve better outcomes.

This strategy enabled:

- minimizing mistakes
- enhancing the efficiency of the system
- maximizing accuracy

8.5 Conclusion

In this chapter, the implementation strategy of the project was discussed along with its timeline. The development of the project was carried out in several phases for effective management.

By employing an organized strategy, the project was effectively developed and improved, leading to an optimal smart traffic signal system.

Chapter – 9

EXPECTED OUTCOMES

9.1 Introduction

The chapter outlines the anticipated output and performance of the proposed traffic management smart system.

The output targets efficiency, decongestion, and balance in traffic flows across various lanes.

9.2 Efficient Traffic Movement

The system aims to enhance traffic movement through junctions.

- The waiting time at signals is minimized
- Traffic movement is more harmonized across various lanes
- The system reduces congestion, particularly during peak hours

It optimizes the use of available roads.

9.3 Adaptability in Signal Timing

Unlike the conventional system, the proposed system adopts adaptable signal timing.

- The period during which signals turn green is determined by current traffic conditions
- Lanes that experience more congestion get priority
- The system regularly updates its signal decisions

This ensures efficient traffic signals operations.

9.4 Fair Allocation of Signal Time

The proposed system features a mechanism of ensuring fairness in signal allocation.

- The waiting time of vehicles in each lane is taken into account
- Nevertheless, no lane is neglected or delayed indefinitely
- Every lane receives a chance to get the green light

9.6 Realistic Traffic Simulation

It offers a realistic simulation of traffic signals.

- Green lane indicates moving lane
- Red lanes indicate halted lanes based on video freeze concept
- The transitions from green to yellow to red signal lights are visible.

It makes it understandable for demonstration purposes.

9.7 Minimized Human Interference

It minimizes manual intervention in controlling traffic lights.

- Traffic lights are automatically controlled
- Manual traffic control is not required
- It constantly monitors and makes decisions

It enhances its effectiveness and performance.

9.8 Scalability and Further Developments

It is scalable and can be expanded for further developments.

- It can be implemented in real-time traffic intersections
- It can integrate with closed-circuit TV cameras
- It can connect with hardware-based traffic controllers

It is appropriate for use in smart cities.

9.9 Conclusion

This chapter provided the anticipated outputs of the project. It seeks to offer efficient and effective traffic management through an intelligent system.

With the integration of Artificial Intelligence, Computer Vision, and multi-lane coordination, the project will bring immense improvement in traffic management systems.

Fig 9.1 Outcome of Proposed System

Chapter – 10

CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

In the current project, a smart and optimized traffic signal system has been developed through the use of Artificial Intelligence and Computer Vision. As compared to conventional fixed-time traffic signals, the system provides a smarter way of functioning through an adaptive algorithm that takes into account various variables depending on real-time situations at the traffic light intersection.

Through the use of videos, this proposed system detects vehicles, estimates their number within the area, and analyzes congestion across all lanes. One difference from simple traffic signals is that it guarantees that one lane gets the green light at a certain time period to maintain proper synchronization between traffic lights.

Another important element of this system is its fairness mechanism, which avoids starving any particular lane while distributing lights according to the estimated traffic conditions. Rather than allocating lights to the most congested lane continuously, waiting times are also taken into consideration.

Occupancy-based estimation further increases the accuracy of the system when the vehicle density makes detection of each vehicle hard to achieve. The use of optimization methods such as frame skipping improves efficiency.

Overall, this system proves the effectiveness of using Artificial Intelligence in traffic light management.

10.2 Future Scope

Even though the existing system works well enough as a simulation, there are certain areas in which it could be improved upon and even enhanced:

- The ability to work in real time, that is, being capable of integrating itself with live CCTV cameras
- The possibility to integrate the algorithm with microcontrollers and traffic signals in general for practical applications
- The ability to detect emergency vehicles such as ambulances, giving them special

priority

- Dataset-dependent machine learning models that are specifically trained on traffic datasets
- An integration with Internet of Things technology to help the process along
- The ability to work with a mobile or web interface for more control over the system
- The ability to process data in the cloud rather than locally

These improvements will help develop the system into a proper smart traffic management tool.

10.3 Summary

The results of this project were shown in this chapter and future improvements were also proposed. This project shows that by using AI and Computer Vision together, it is possible to address practical problems.

If the project will be developed even more, it may become an important addition to traffic systems

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